THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–8: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature, Measurement of Time and Motion, Life Processes in Animals** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** sexually mature	Maturity.
2	**(d)** hot to cold	Always.
3	**(c)** second	Second.
4	**(c)** mouth	Ingestion.
5	**(c)** boys	Puberty.
6	**(a)** sinks	Heavier.
7	**(c)** distance / time	Speed formula.
8	**(a)** lungs	Lungs.
9	**(c)** parents and teachers	Trusted adults.
10	**(c)** solid	Solid.
11	**(a)** uniform	Uniform.
12	**(d)** heart	Heart.
13	**(d)** 11 and 19 years	Teen.
14	**(c)** radiation	Radiation.
15	**(b)** 60 seconds	60 s.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Time of sexual maturity (1); Between 10 and 14 years (1).	1+1
17	Conduction (1); Convection and radiation (1).	1+1
18	Distance per unit time (1); Unit is m/s (1).	1+1
19	Breaking food into simple parts (1); In the alimentary canal (1).	1+1
20	Thyroid (1); Makes thyroxine (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Height increases (1); Voice deepens in boys (1); Body hair grows (1).	1+1+1
22	Spoon in hot tea (1); Iron rod in fire (1); Hot pan handle (1).	1+1+1
23	Second (1); Minute (1); Hour (1).	1+1+1
24	Mouth (1); Stomach (1); Intestine (1).	1+1+1
	OR: Tobacco (1); Alcohol (1); Drugs (1).	1+1+1
25	Cotton shirt (1); Light colours (1); Loose clothes (1).	1+1+1
	OR: Distance = speed x time (1); 40 x 2 (1); 80 km (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Height (1); Voice (1); Hair (1); Sweat glands (1); Mood (1).	1+1+1+1+1
	OR: Heat moves in a solid (1); Spoon in hot tea (1); Particle to particle (1); Metals conduct (1); Wood does not (1).	1+1+1+1+1
27	Speed = distance / time (1); 120/20 (1); 6 (1); m/s (1); 6 m/s (1).	1+1+1+1+1
	OR: Ingestion (1); Digestion (1); Absorption (1); Assimilation (1); Egestion (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–8: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature, Measurement of Time and Motion, Life Processes in Animals** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** teeth	Teeth.
2	**(c)** girls and women	Females.
3	**(d)** in the day	From sea to land.
4	**(a)** m/s	Metre per second.
5	**(b)** nose	Nostrils.
6	**(c)** friends	Same age.
7	**(b)** fluid	Liquid or gas.
8	**(d)** non-uniform	Non-uniform.
9	**(b)** blood vessels	Arteries, veins.
10	**(d)** chemical messengers	Control body.
11	**(a)** conduction	Solid.
12	**(b)** 60 minutes	60 min.
13	**(d)** oesophagus	Carries food.
14	**(b)** spinach	Iron.
15	**(b)** at night	From land to sea.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Iron (1); Copper (1).	1+1
17	Time for one oscillation (1); In seconds (1).	1+1
18	Moistens food (1); Digests starch (1).	1+1
19	For thyroxine (1); To avoid goitre (1).	1+1
20	Wind from sea (1); During the day (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Amylase (1); Pepsin (1); Lipase (1).	1+1+1
22	Oil glands active (1); Dirt blocks pores (1); Hormones (1).	1+1+1
23	Double walls (1); Vacuum (1); Cork (1).	1+1+1
24	Second hand (1); Fan blades (1); Escalator (1).	1+1+1
	OR: Clean air (1); Exercise (1); No smoking (1).	1+1+1
25	Pituitary (1); Thyroid (1); Pancreas (1).	1+1+1
	OR: Boiling water (1); Sea breeze (1); Room heater (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Chemical messengers (1); Made by glands (1); Travel in blood (1); Control growth (1); Control mood (1).	1+1+1+1+1
	OR: Hot tea (1); Cold drinks (1); Soup (1); Travel (1); Picnics (1).	1+1+1+1+1
27	18 x 5/18 (1); 5 m/s (1); 10 x 18/5 (1); 36 km/h (1); Use the factor (1).	1+1+1+1+1
	OR: Two kidneys (1); Bean shaped (1); Filter blood (1); Make urine (1); Remove waste (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–8: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature, Measurement of Time and Motion, Life Processes in Animals** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 50 km/h	100/2.
2	**(a)** carbon dioxide	Waste gas.
3	**(b)** say no	Refuse.
4	**(c)** waves	Rays.
5	**(c)** non-uniform motion	Stops and starts.
6	**(a)** haemoglobin	Pigment.
7	**(c)** testes	In males.
8	**(a)** convection	Fluid movement.
9	**(b)** 24 hours	24 h.
10	**(a)** stomach	Pepsin.
11	**(d)** goitre	Iodine.
12	**(d)** more heat	Absorb.
13	**(b)** 5 m/s	10/2.
14	**(b)** diaphragm	Breathing.
15	**(b)** change often	Mood swings.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Unequal distance in equal time (1); Changing speed (1).	1+1
17	White blood cells (1); Fight germs (1).	1+1
18	Voice deepens (1); Beard grows (1).	1+1
19	Heat transfer in solids (1); Particle to particle (1).	1+1
20	Second (1); Symbol s (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Sundial (1); Water clock (1); Quartz clock (1).	1+1+1
22	Nose (1); Windpipe (1); Lungs (1).	1+1+1
23	Make sperms (1); Make testosterone (1); Cause male changes (1).	1+1+1
24	Sweater (1); Woollen cap (1); Jacket (1).	1+1+1
	OR: Time = distance / speed (1); 150/50 (1); 3 h (1).	1+1+1
25	Nose (1); Lungs (1); Diaphragm (1).	1+1+1
	OR: Growth (1); Mood (1); Reproduction (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Pituitary: growth hormone (1); Thyroid: thyroxine (1); Pancreas: insulin (1); Testes: testosterone (1); Ovary: oestrogen (1).	1+1+1+1+1
	OR: Fluid moves (1); Hot water rises (1); Cool water sinks (1); Currents form (1); Boiling water (1).	1+1+1+1+1
27	Sundial (1); Water clock (1); Sand timer (1); Pendulum clock (1); Quartz watch (1).	1+1+1+1+1
	OR: Mouth (1); Oesophagus (1); Stomach (1); Small intestine (1); Large intestine and anus (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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